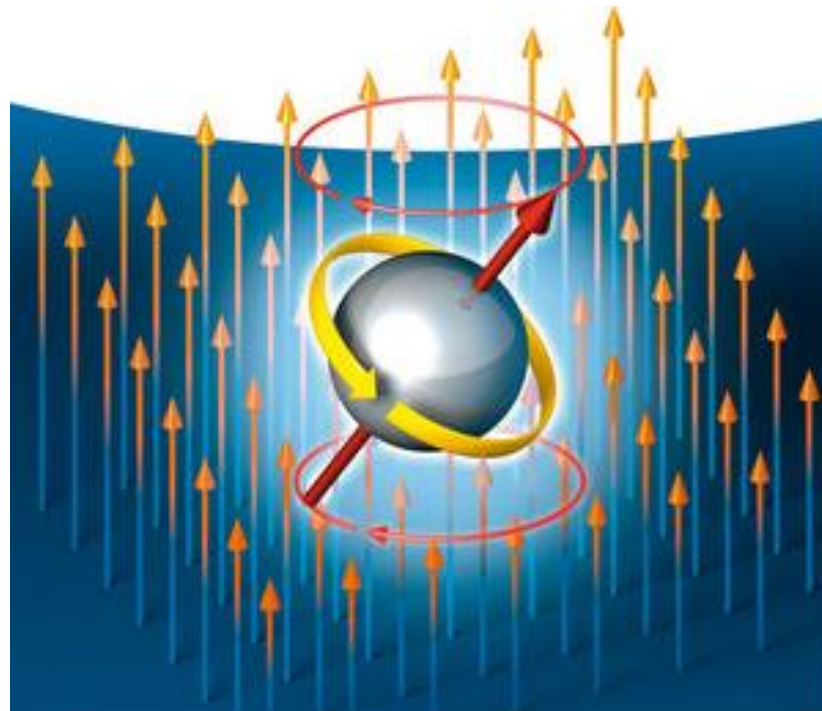


Spectroscopy

Nuclear Magnetic Resonance Spectroscopy

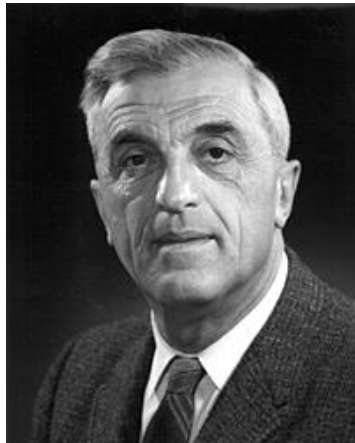


Nuclear Magnetic Resonance (NMR)

Nuclear magnetic resonance (NMR) is a physical phenomenon in which nuclei in a magnetic field absorb and re-emit electromagnetic radiation.

Nobel Prize for Physics in 1952

Felix Bloch, Stanford University, 1945



Edward Purcell, Harvard University, 1945



“Development of new methods for nuclear magnetic precision measurements”

NMR Spectroscopy has become an indispensable tool for chemists, physicists and biologists...

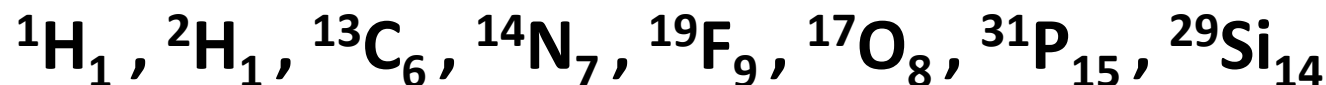
- ***Molecular structural determination***
- ***Study of molecular dynamics***
- ***Characterization of molecular materials***
- ***Structure of protein's and microorganisms etc.***

Theoretical Principles

1. Nuclear Spin

- No. of neutrons & No. of protons are both even - No Nuclear Spin
- No. of neutrons & No. of protons are both odd - Integer Spin
(1, 2, 3 etc.)
- No. of neutrons or No. of protons odd - Half-integer Spin
(1/2, 3/2, 5/2 etc.)x

NMR Active Nuclei



Classification of Nuclei

Group 1 - Nuclei with both At No. and No. of neutrons even
eg. ^{12}C , ^{16}O , ^{18}O , ^{32}S

Group 2 - Nuclei with Both At.No. and No. of neutrons odd
eg. ^2H ($I = 1$), ^{10}B ($I = 3$), ^{14}N ($I=1$)

**Group 3 - Nuclei with Even At.No and Odd Neutrons or
*vice versa***
eg. ^1H , ^{11}B (**$I=3/2$**), ^{13}C , ^{15}N , ^{17}O (**$I=5/2$**), ^{19}F , ^{29}Si , ^{31}P

Magnetic Properties of Selected Nuclei

Isotope	Abundance	Z (At.No)	N (Neut)	A (Mass No)	I	γ
neutron	-	0	1	1	1/2	-183.26
^1H	99.985	1	0	1	1/2	267.512
^2H	0.015	1	1	2	1	41.063
^{13}C	1.108	6	7	13	1/2	67.264
^{17}O	0.037	8	9	17	5/2	-36.27
^{31}P	100	15	16	31	1/2	108.29
^{29}Si	4.70	14	15	29	1/2	-53.142

2. Nuclear Spin States

No. of spin states = $2I + 1$ where I is the nuclear spin
 $+I, (I-1), \dots, (-I+1), -I$

H - $I = \frac{1}{2}$; Spin states: $-1/2, +1/2$

Cl - $I = 3/2$; Spin States: $-3/2, -1/2, +1/2, +3/2$

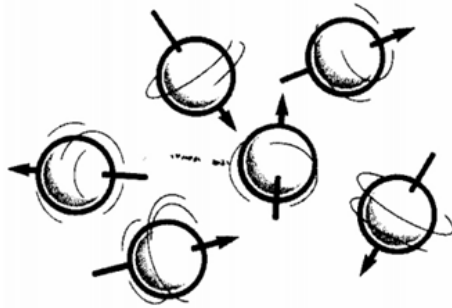
SPIN QUANTUM NUMBERS OF SOME COMMON NUCLEI

Element	^1_1H	^2_1H	$^{12}_6\text{C}$	$^{13}_6\text{C}$	$^{14}_7\text{N}$	$^{16}_8\text{O}$	$^{17}_8\text{O}$	$^{19}_9\text{F}$	$^{31}_{15}\text{P}$	$^{35}_{17}\text{Cl}$
Nuclear spin quantum number	$\frac{1}{2}$	1	0	$\frac{1}{2}$	1	0	$\frac{5}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{2}$
Number of spin states	2	3	0	2	3	0	6	2	2	4

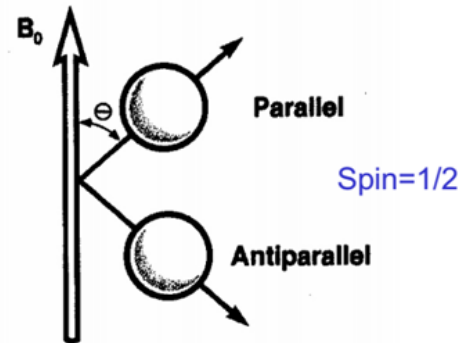
Where does the NMR signal come from?

The Zeeman Effect

- When inserted in a magnetic field (B_0) nuclei that possess *spin* align themselves according to their energy states.
 - This effect on their alignment is called the *Zeeman Effect*.



No B_0



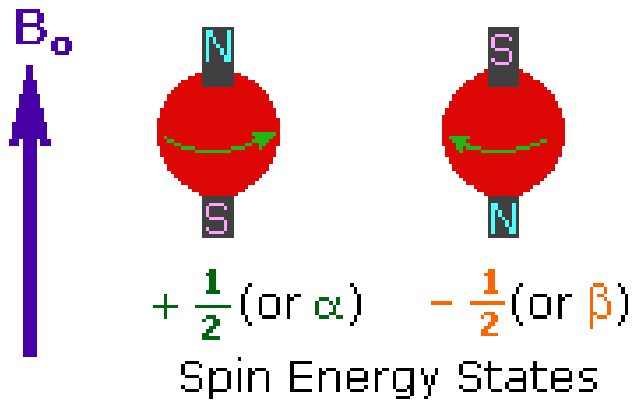
B_0

Pieter Zeeman: 1902 Nobel Prize in Physics with Hendrik Lorentz for his discovery of the 'Zeeman effect'



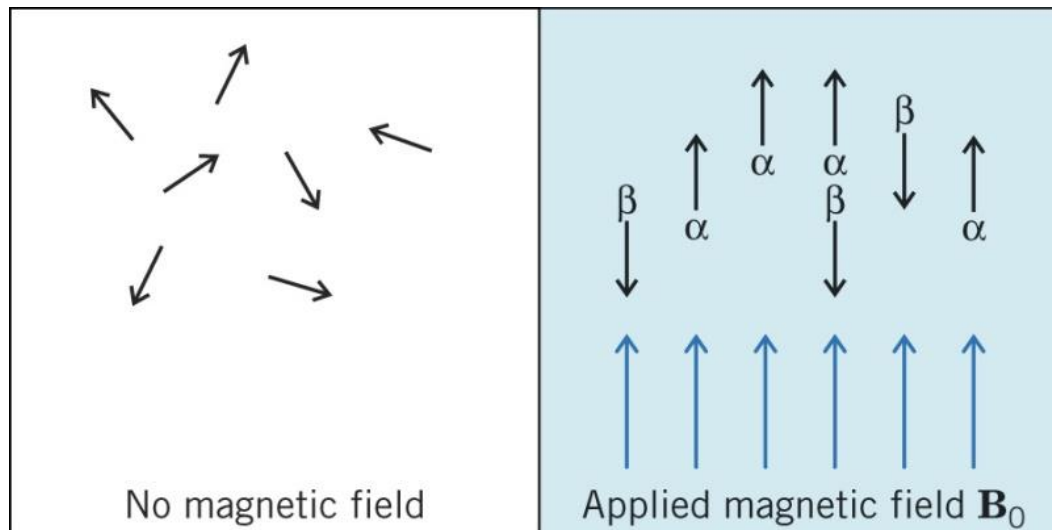
3. Nuclear Magnetic Moments

The **nuclear magnetic moment** is the **magnetic moment** of an atomic nucleus and arises from the spin of the protons and neutrons.



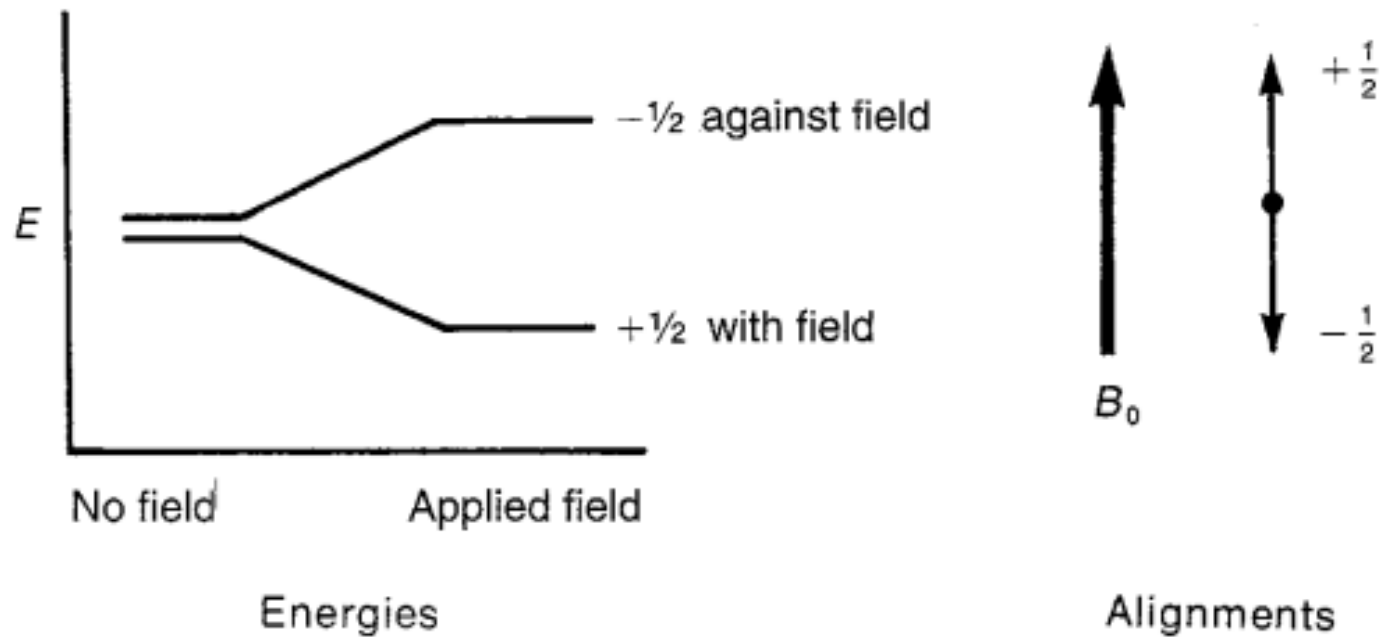
In an applied magnetic field, all protons have their magnetic moments either aligned with the field or opposed to it.

In the absence of an applied field, these protons behave like bar magnets that are randomly oriented.



- The spin states are not of equivalent energy
- The nucleus being a charged particle with a spin generates a magnetic moment μ
- For H-atom the two magnetic moments generated by the two spin states are aligned with the applied magnetic field or opposed to it
- The spin state $+1/2$ is of lower energy (aligned with the field and $-1/2$ is of higher energy (aligned against the applied field)

Spin States of proton in the absence and presence of an applied magnetic field



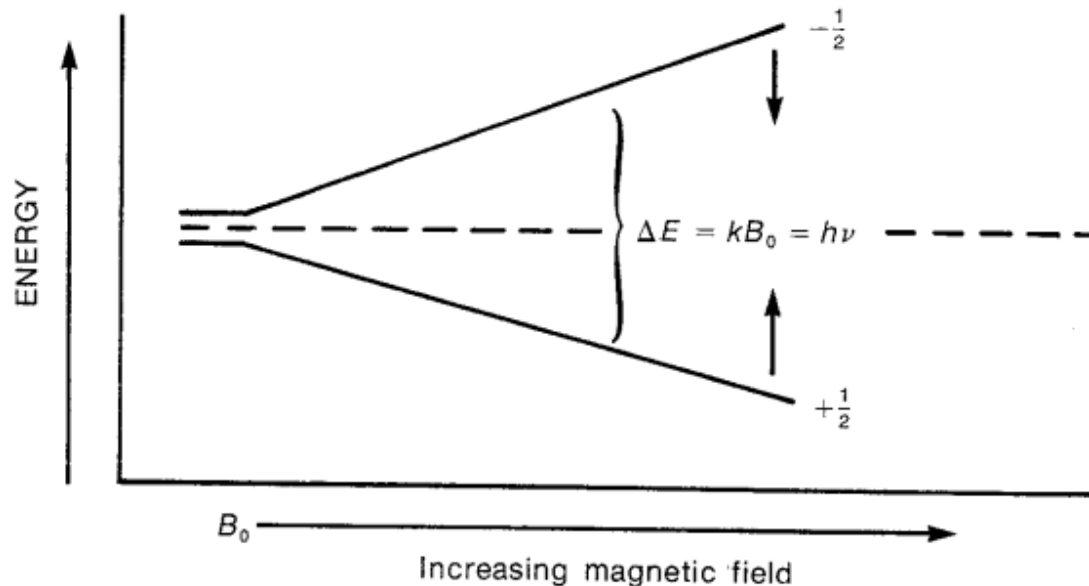
4. Absorption of energy

Nuclear Magnetic Resonance Phenomenon occurs when nuclei aligned with the applied magnetic field are induced to absorb energy, and change their spin orientation with respect to the applied field

Energy absorbed is quantized = energy difference between the two states

$$\text{Energy}_{\text{abs}} = E_{-1/2 \text{ state}} - E_{+1/2 \text{ state}} = h\nu$$

The spin state-energy separation as a function of the applied magnetic field; B_0



The spin state-energy separation as a function of the applied magnetic field; B_0

The stronger the applied magnetic field, the greater the energy difference

$$\Delta E = f(B_0)$$

Magnitude of energy separation depends on the particular nucleus involved (magnetogyric ratio – ratio of the magnetic moment to angular momentum)

$$\Delta E = f(\gamma B_0) = h\nu$$

Angular momentum of the nucleus is quantized in units of $h/2\pi$

$$\Delta E = \gamma (h/2\pi) B_0 = h\nu$$

$$\nu = (\gamma/2\pi) B_0$$

$$\nu = (\gamma/2 \pi) B_0$$

$$\gamma \text{ } ^1\text{H} = 267.54 \text{ rad/T}; B_0 = 1\text{T}$$

$$\nu = 267.54 \times 1 / 2 \times 3.14 = 42.6 \text{ MHz}$$

Frequency of radiation	Field strength B_0
42.6 MHz	1 T (10,000 Gauss)
60.0 MHz	1.4 T (14,000 Gauss)
100 MHz	2.35 T
300 MHz	7.05 T
500 MHz	11.1 T

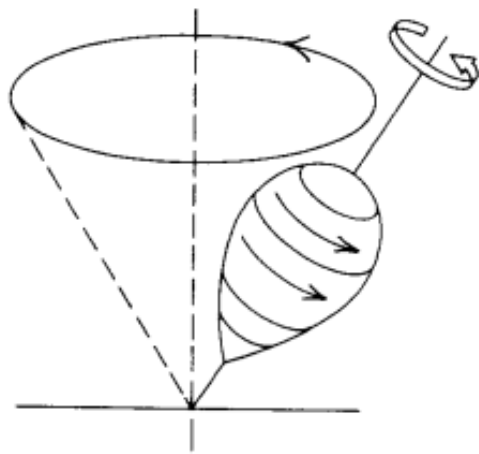
5. Nuclear Magnetic Resonance

When placed in a magnetic field and irradiated with radio frequency energy, these nuclei absorb energy at frequencies based on their chemical environments

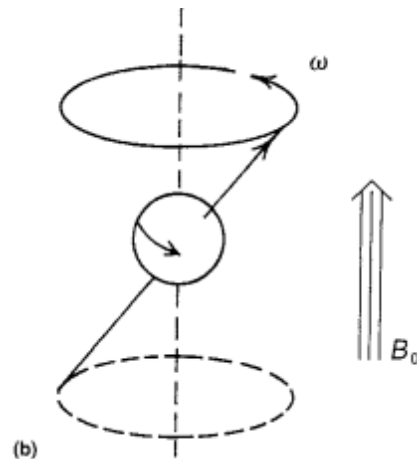
In the applied magnetic field, nuclei begin to precess

Angular Frequency ω or Precessional Frequency or Larmor Frequency

Precessional Frequency $\propto B_0$



A top precessing in earth's gravitational field



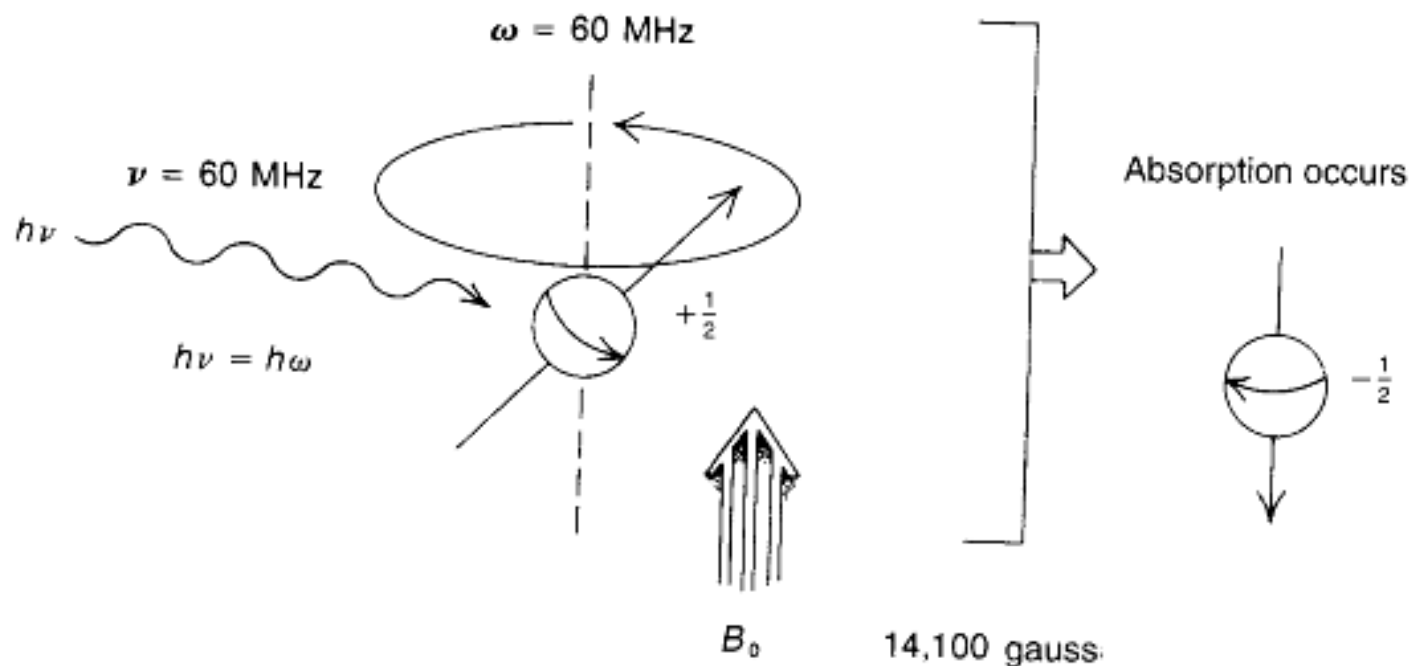
Precession of a spinning nucleus in an applied magnetic field

Resonance Phenomenon

- Precession of the nucleus generates an oscillating electric field of the same frequency
- When a radiofrequency wave of this frequency is supplied to the precessing nucleus, energy can be absorbed

Frequency of the oscillating electric field component of incoming radiation
= frequency of the electric field generated by the precessing nucleus

---- energy transfer to the nucleus -- spin change ---- resonance



- The nucleus in the applied magnetic field precess with angular frequency ω –Larmor Frequency
- Larmor Frequency is directly proportional to field strength
- Precession generates an oscillating electric field of same frequency
- If radioenergy of same frequency is supplied to the nucleus Energy can be transferred from the incoming radiation to the nucleus, causing a spin change -- Resonance - the nucleus is in resonance with the incoming radiofrequency.

Recap

- Nuclei with spin behaves like a bar magnet and has nuclear magnetic moment
- In the absence of a magnetic field, the spin states are degenerate – of equal energy. The spins cancel each other and will have zero magnetic moment
- When placed in an external magnetic field, the spin states lose their degeneracy and get separated and the nuclei begin to precess with a frequency which is directly proportional to the external magnetic field.

$$\Delta E = \gamma \left(\frac{h}{2\pi} \right) B_0 = h\nu$$

- When radiofrequency equal to the precessional frequency is applied, absorption of Energy takes place

The NMR Spectrometer

- Niobium-tin-copper clad coil wound like a spool of thread. The current runs through this coil, creating the magnetic field.
- This coil is submerged in liquid helium (boiling point -452.1°F , -268.9°C , 4°K – Brrr!!!!) .
- Liquid He chamber surrounded by liquid nitrogen (boiling point -320.5°F , -195.8°C , 77°K).
- Sample and spinner lowered using air from the top, down through the bore, until it nests in the top of the probe.

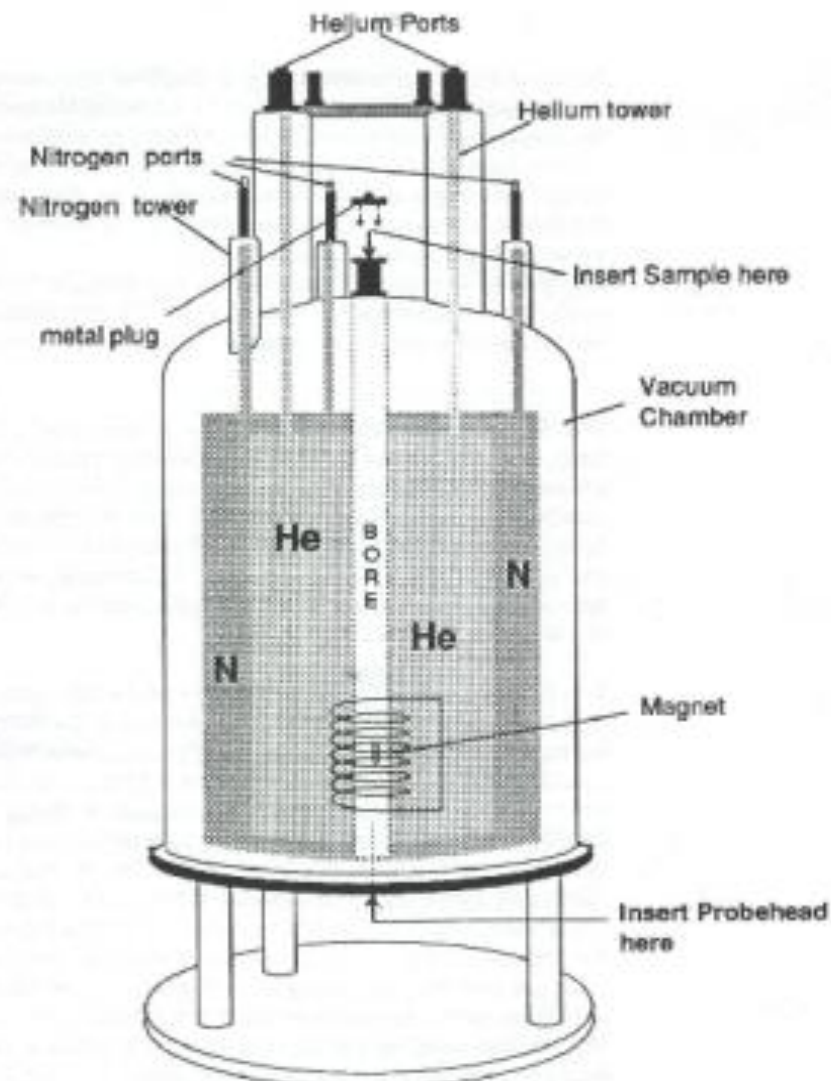
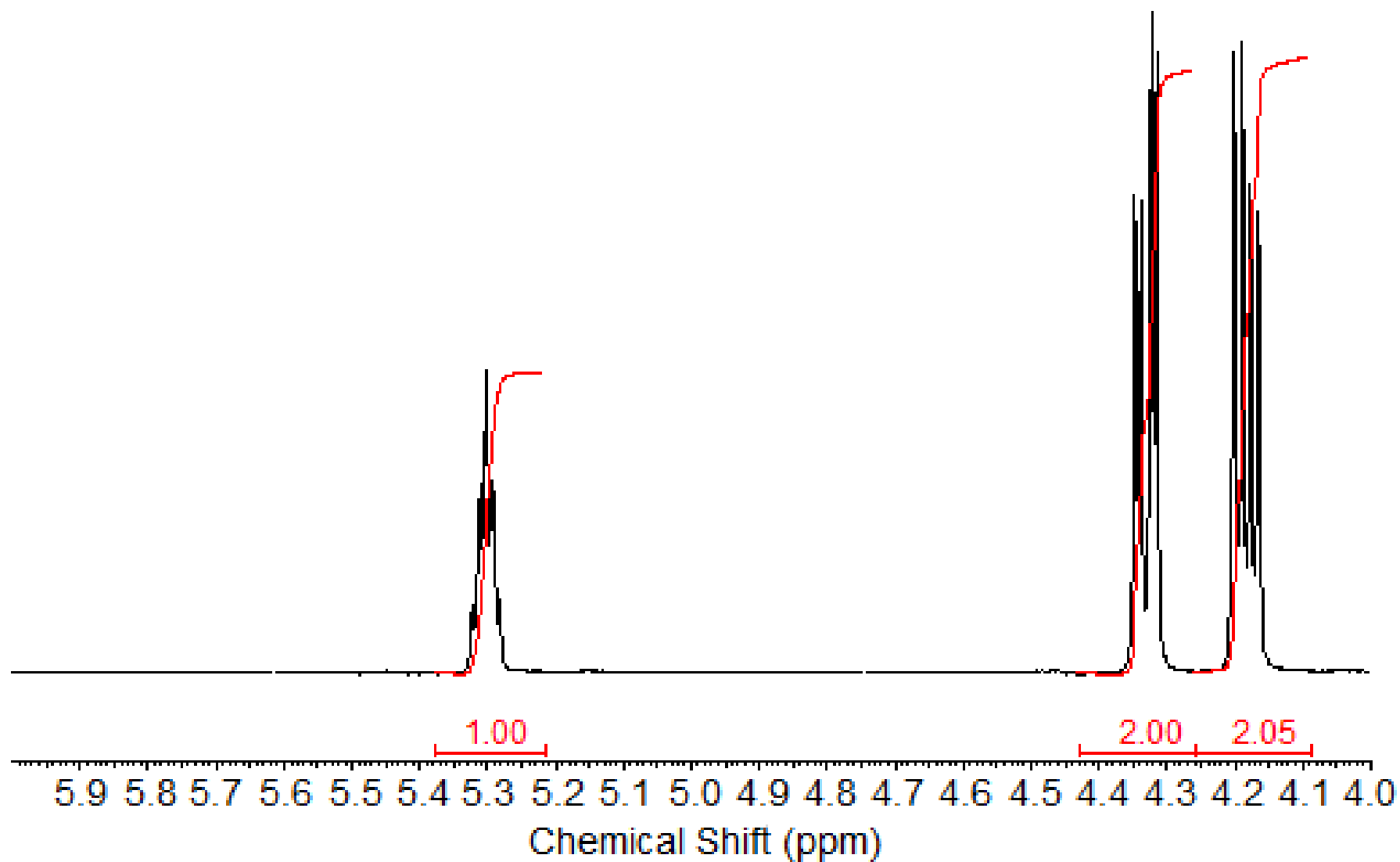


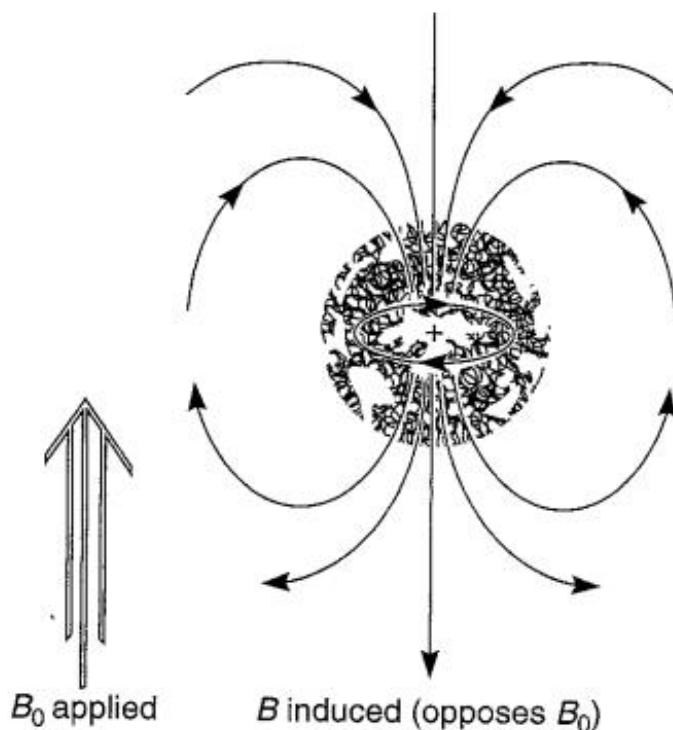
Figure 3.B Magnet and dewar

^1H NMR Spectrum



Shielding Effect

Protons in different environment are shielded differently by the electrons surrounding them.

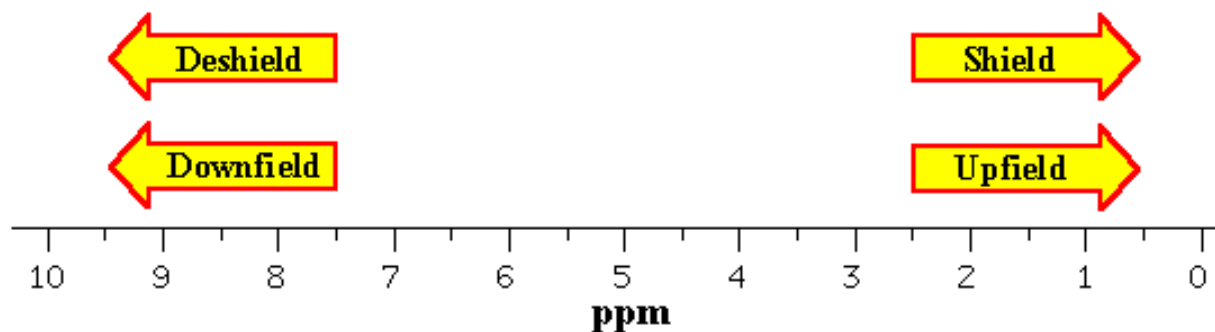


Diamagnetic anisotropy: Diamagnetic shielding of a nucleus caused by the circulation of valence electrons.

Chemical shift

S-electrons - spherical symmetry - secondary magnetic field opposing the magnetic field - nuclear shielding - upfield shift (diamagnetic shift)

P-Orbitals - no spherical symmetry - produces comparatively large magnetic fields at the nucleus - deshielding - (paramagnetic shift)



Chemical shift

The chemical shift is the resonant frequency of a nucleus relative to a standard in a magnetic field

Standard: Tetramethylsilane (TMS)

$$\delta = \frac{[\nu_o(\text{H}) - \nu_o(\text{TMS})]}{[\text{Spectrometer Frequency in MHz}]}$$

Larmor precession frequency (ν_o)

Exact Resonance Frequency depends on the electronic environment of the nucleus

$$B_{\text{eff}} = B_o - \sigma B_o$$

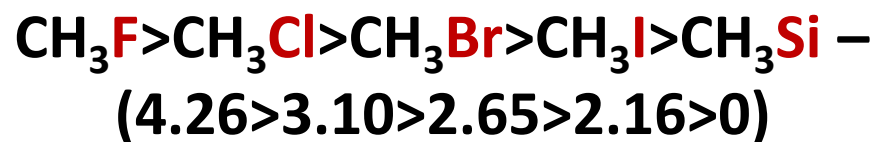
σ = shielding constant

*Stronger the nucleus is shielded, larger σ is, smaller the B_{eff} becomes
For constant frequency the applied field B_o must be larger for resonance*

Factors Influencing Chemical Shift

➤ Electronegativity

The greater the electronegativity of the atoms attached to carbon atom, the more the deshielding, thus greater is the chemical shift of the protons



R_4N^+ - Highly deshielded (high δ value)

Carbanionic centres - Highly shielded (low δ value)

A. Electronegativity Effects

TABLE 3.4

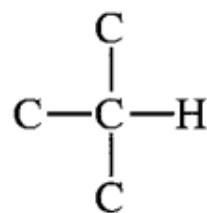
APPROXIMATE CHEMICAL SHIFT RANGES (PPM) FOR SELECTED TYPES OF PROTONS^a

$R-CH_3$	0.7 – 1.3	$R-\overset{\overset{ }{\text{N}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	2.2 – 2.9
$R-CH_2-R$	1.2 – 1.4	$R-S-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	2.0 – 3.0
R_3CH	1.4 – 1.7	$I-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	2.0 – 4.0
$R-\overset{\overset{ }{\text{C}}}{\text{C}}=\overset{\overset{ }{\text{C}}}{\text{C}}-\overset{\overset{ }{\text{C}}}{\text{C}}-H$	1.6 – 2.6	$Br-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	2.7 – 4.1
$R-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H, H-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	2.1 – 2.4	$Cl-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	3.1 – 4.1
$RO-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H, HO-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	2.1 – 2.5	$R-\overset{\overset{\text{O}}{\parallel}}{\underset{\underset{\text{O}}{\parallel}}{\text{S}}}-O-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	ca. 3.0
$N\equiv C-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	2.1 – 3.0	$RO-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H, HO-\overset{\overset{ }{\text{C}}}{\underset{\underset{ }{ }}{\text{C}}}-H$	3.2 – 3.8

B. Hybridization effects

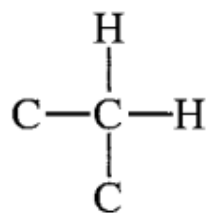
sp^3 hydrogens

Aliphatic region



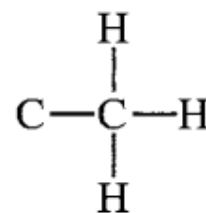
3°

>



2°

>



1°

>



Strained ring

2

1

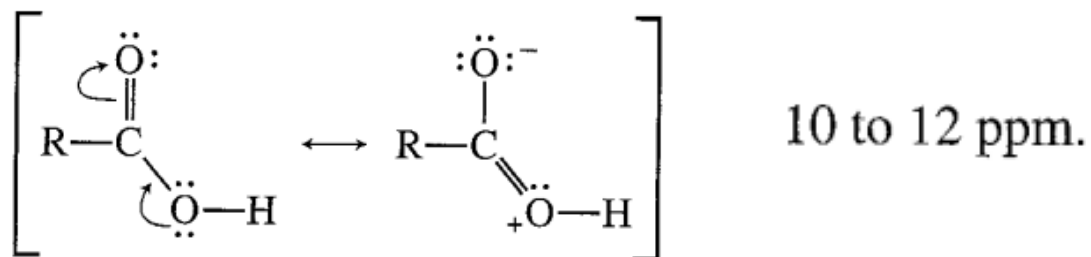
0 δ

sp^2 hydrogens

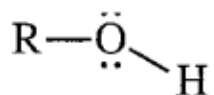
Hydrogens attached to the sp^2 carbons have greater chemical shift than hydrogens on aliphatic hydrogens on sp^3 carbon

C. Acidic and exchangeable hydrogens;; hydrogen bonding

Acidic protons

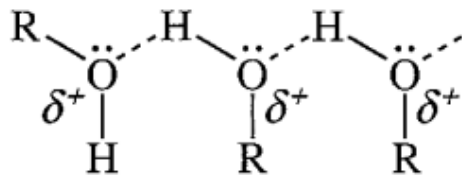


Hydrogen bonding and exchangeable hydrogens



0.5–1.0 ppm

Free (dilute solution)



Hydrogen-bonded (concentrated solution)

TYPICAL RANGES FOR PROTONS WITH VARIABLE CHEMICAL SHIFT

Acids	RCOOH	10.5–12.0 ppm
Phenols	ArOH	4.0–7.0
Alcohols	ROH	0.5–5.0
Amines	RNH ₂	0.5–5.0
Amides	RCONH ₂	5.0–8.0
Enols	CH=CH–OH	>15

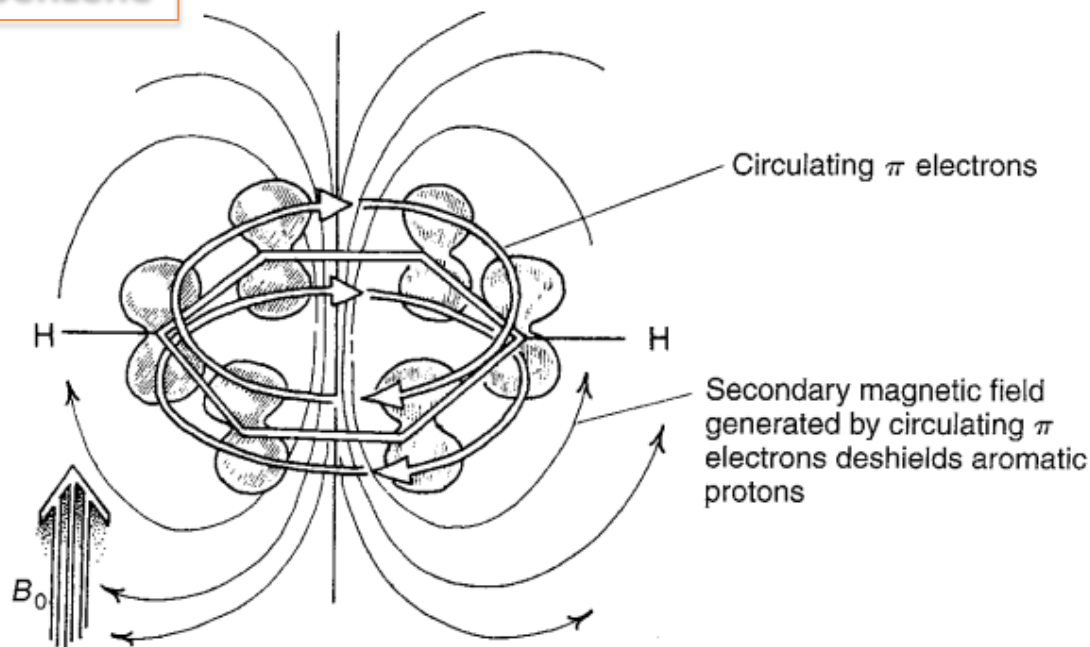
➤ Anisotropic Effects

- π Electrons circulate under the influence of applied magnetic field and Create a secondary field which can either oppose or augment it.
- Causes shift to higher frequency (downfield shifts or diamagnetic shift)
Or
To low frequency (upfield shift or paramagnetic shift) – Anisotropic Effect

Magnetic Anisotropy in Benzene and Acetylene

Anomalous shift in resonance values is due to the presence of an unsaturated system in the vicinity of the proton in question.

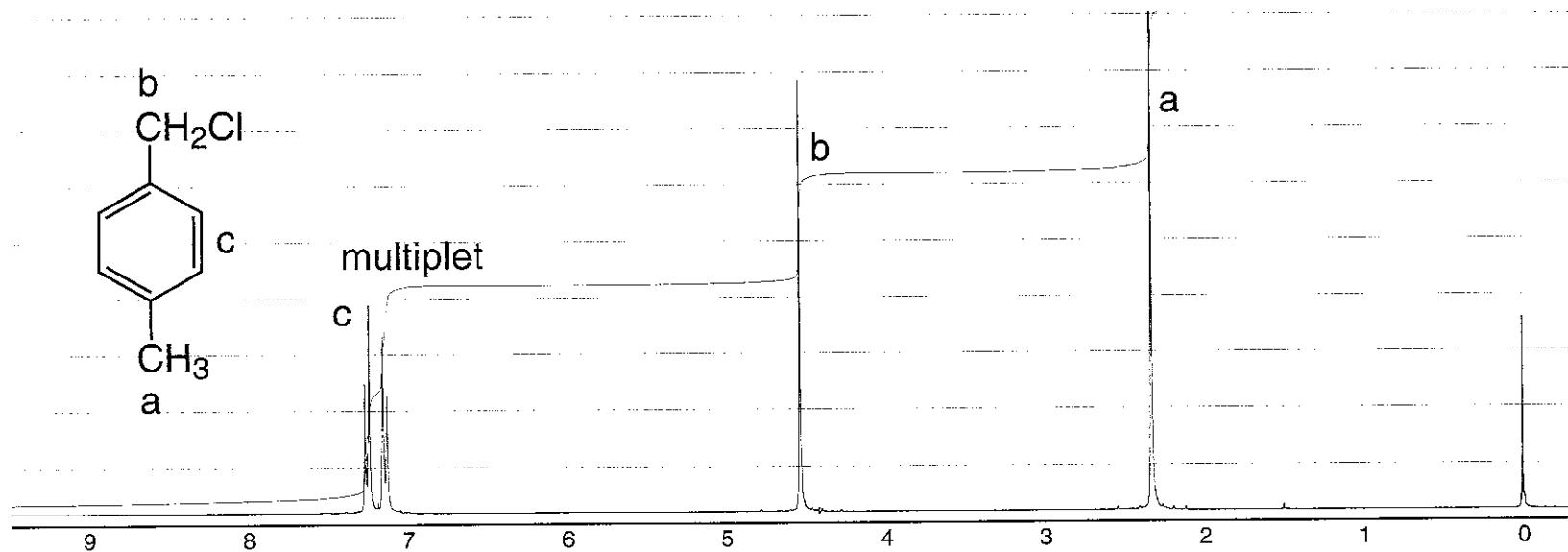
Diamagnetic anisotropy in benzene



The protons on the periphery of the benzene ring happen to lie in a deshielding region of the anisotropic field; this gives the benzene protons a δ value that is greater than expected

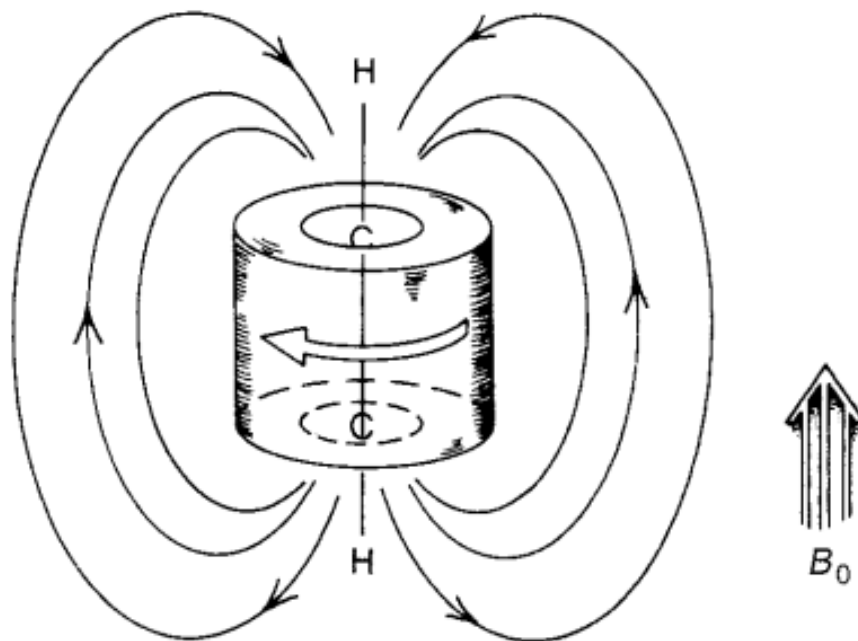


Example.....

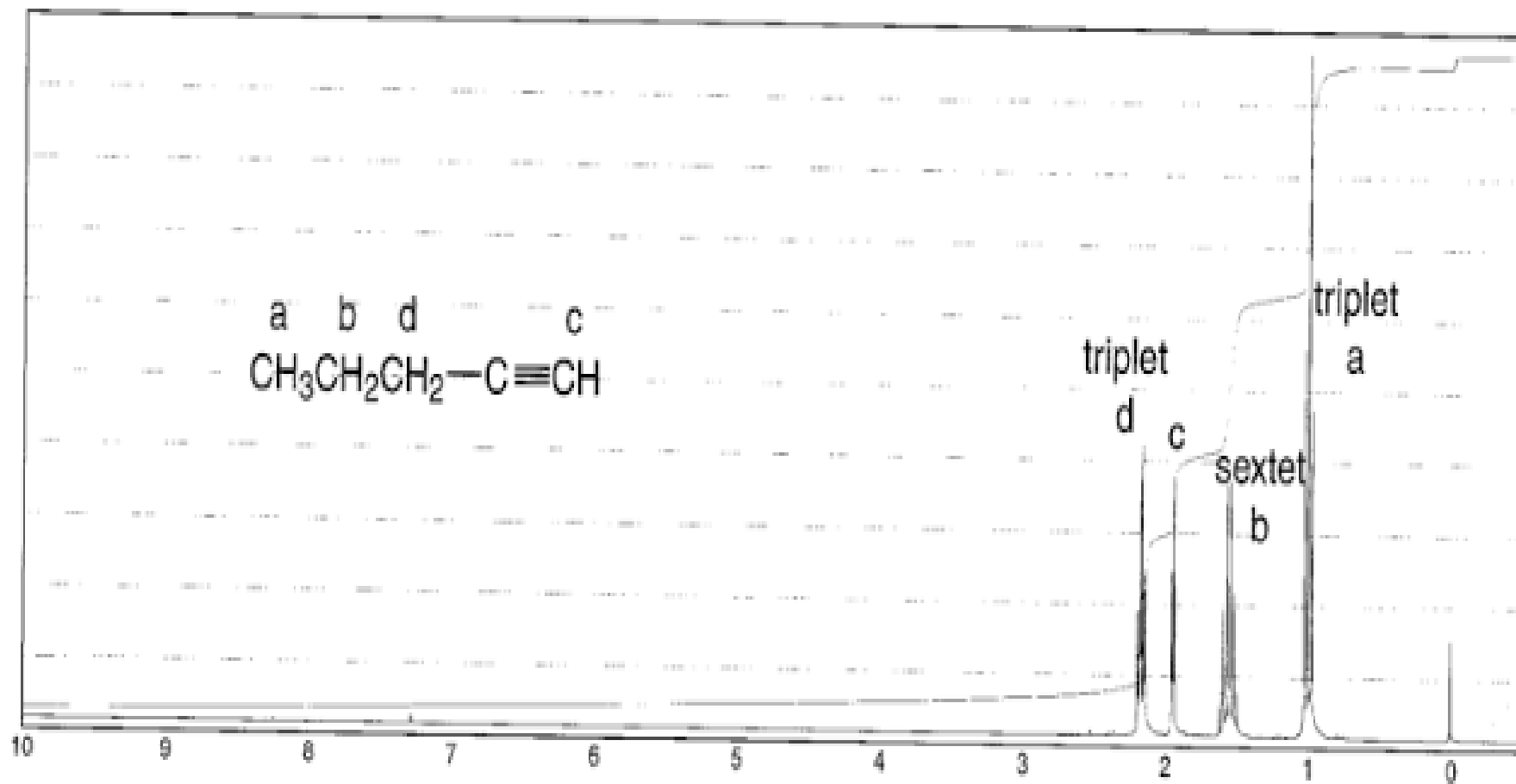


^1H NMR spectrum of α -chloro-*p*-xylene

Diamagnetic anisotropy in acetylene



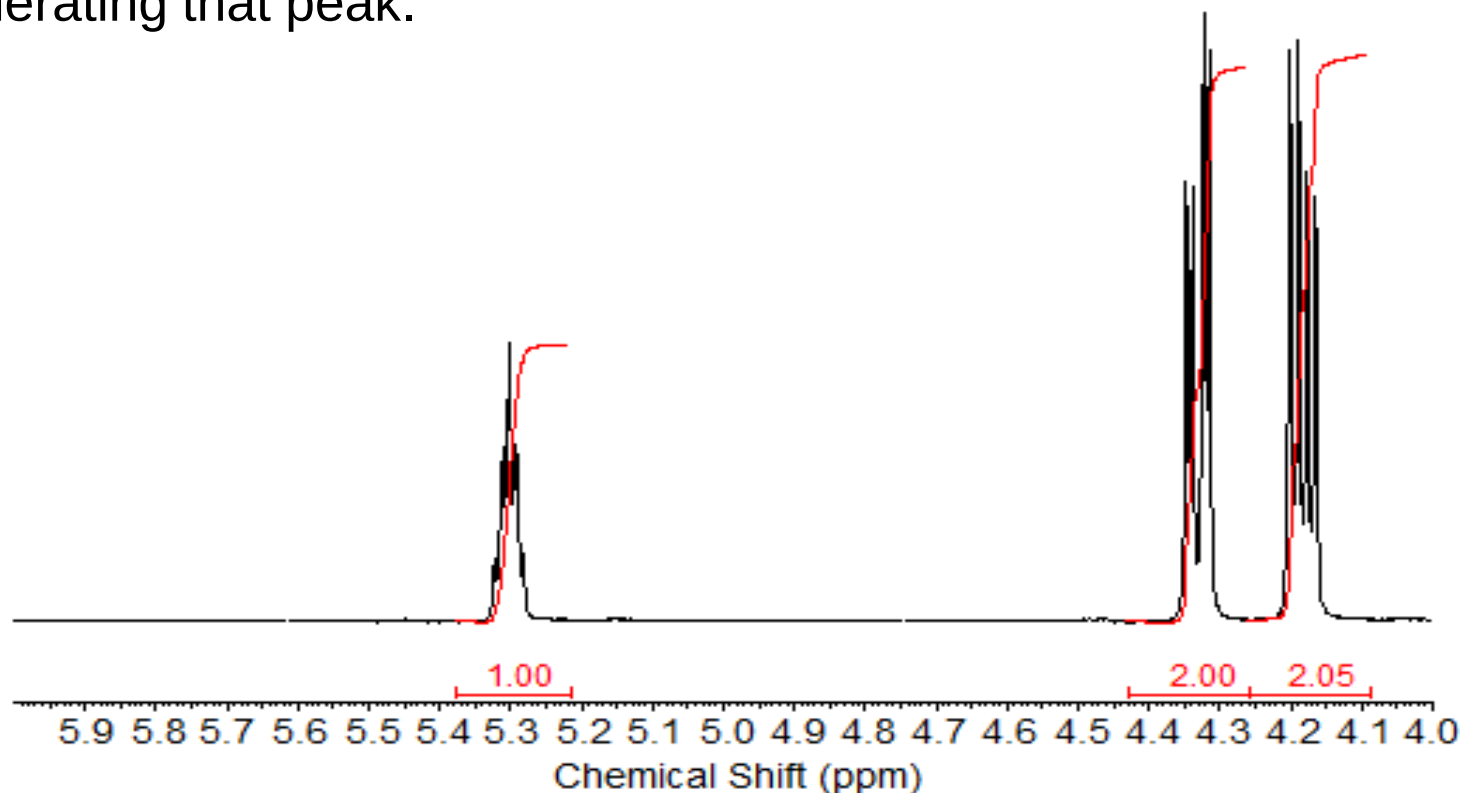
In acetylene, the magnetic field generated by induced circulation of the p electrons has a geometry such that the acetylenic hydrogens are shielded. Hence they have a resonance at a higher field than expected.



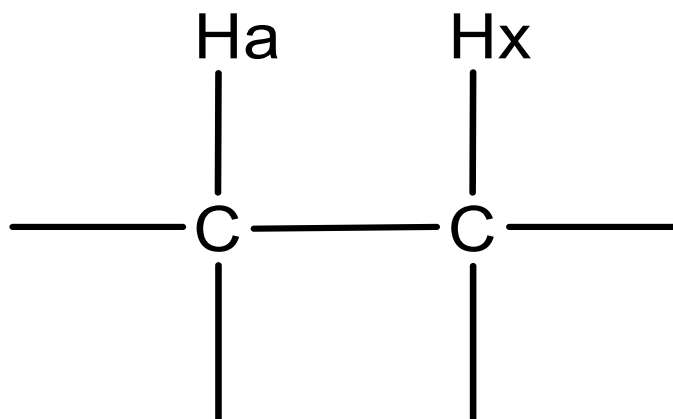
^1H spectrum of 1-pentyne

Integrals and Integration

- NMR spectrum not only distinguishes the different types of protons in a molecule, but also reveals how many of each type are contained within the molecule.
- The area under each peak is proportional to the number of hydrogens generating that peak.



Spin-spin splitting (n+1) rule



Multiplicity of lines is related to the number of neighbouring protons

n+1 rule: Each type of proton “senses” the number of equivalent protons (n) on the carbon atom(s) next to the one to which it is bonded, and its resonance peak is split into (n+1) components.

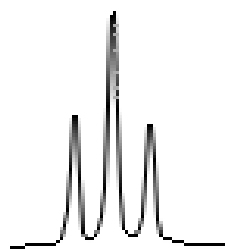


Spin-spin splitting (n+1) rule

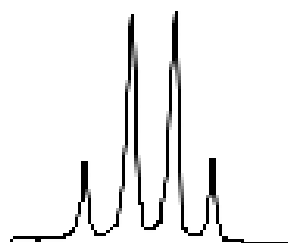
Pascal's Triangle

The intensity ratios of multiplets derived from the n+1 rule follow the entries in the mathematical mnemonic device called Pascal's triangle

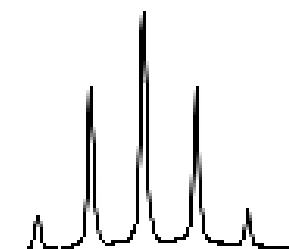
Singlet	1							
Doublet	1		1					
Triplet	1		2	1				
Quartet	1		3	3	1			
Quintet	1		4	6	4	1		
Sextet	1		5	10	10	5	1	
Septet	1		6	15	20	15	6	1



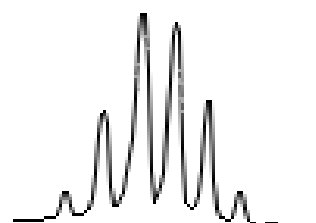
triplet $n = 2$
1:2:1



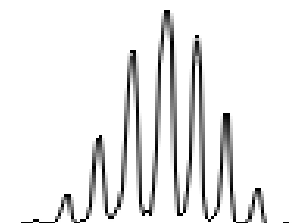
quartet $n = 3$
1:3:3:1



pentet $n = 4$
1:4:6:4:1

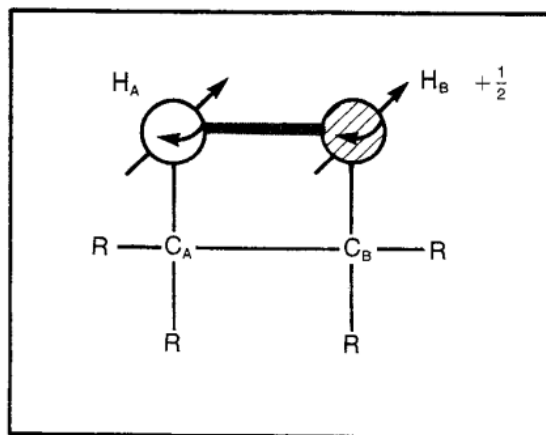


sextet $n = 5$
1:5:10:10:5:1

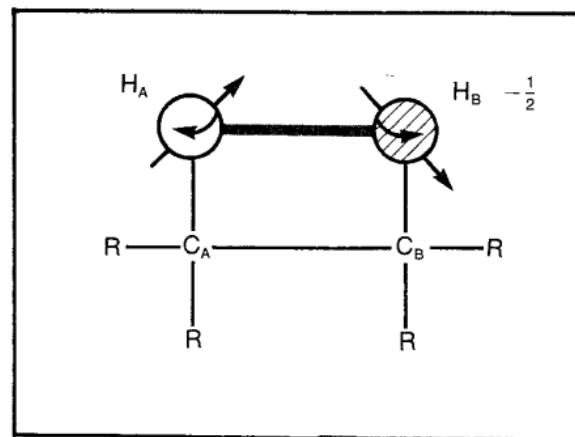


nonet $n = 8$
1:6:15:20:15:6:1

The origin of spin-spin splitting

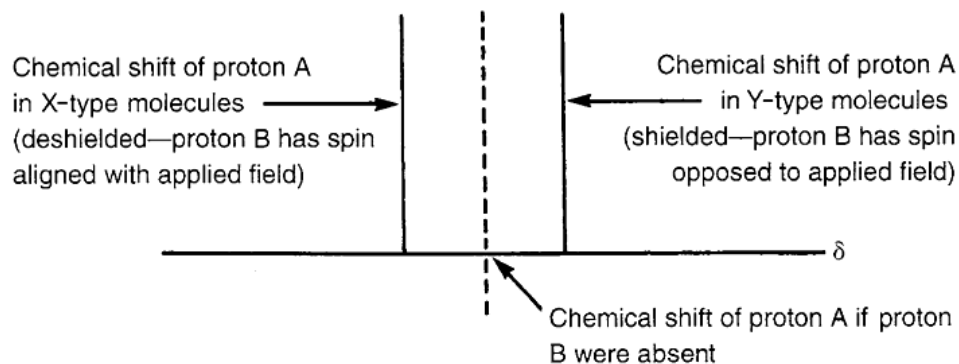


X-type molecule



Y-type molecule

Spin-spin splitting arises because hydrogens on adjacent carbon atom can “**sense**” one another



COUPLING CONSTANTS (J)

- The distance between peaks in a multiplet is called coupling constant ' J '
- ' J ' is a measure of how strongly a nucleus is affected by the spin states of its neighboring nuclei.
- ' J ' is expressed in Hertz (Hz).
- ' J ' of a group of protons that split one another must be identical.

